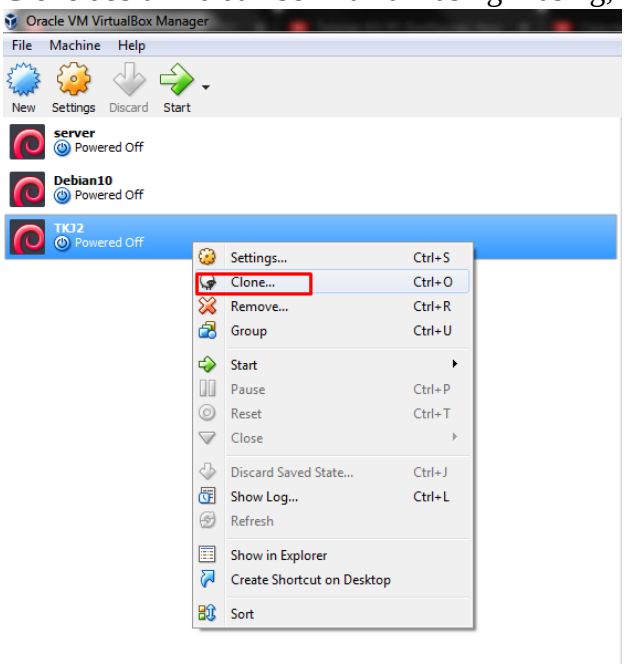
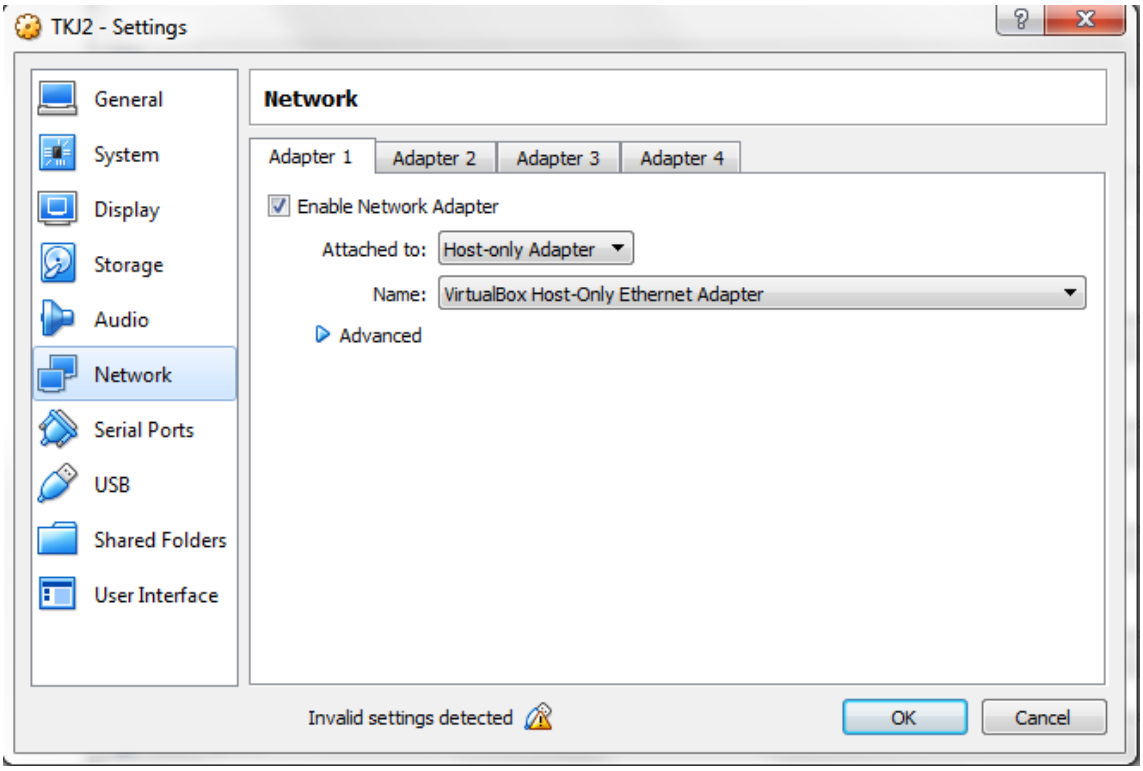
KONFIGURASI DEBIAN SERVER 10

1. Clone debian 10 dan beri nama masing-masing, format: Debian XIITKJ(kelas)(nama)

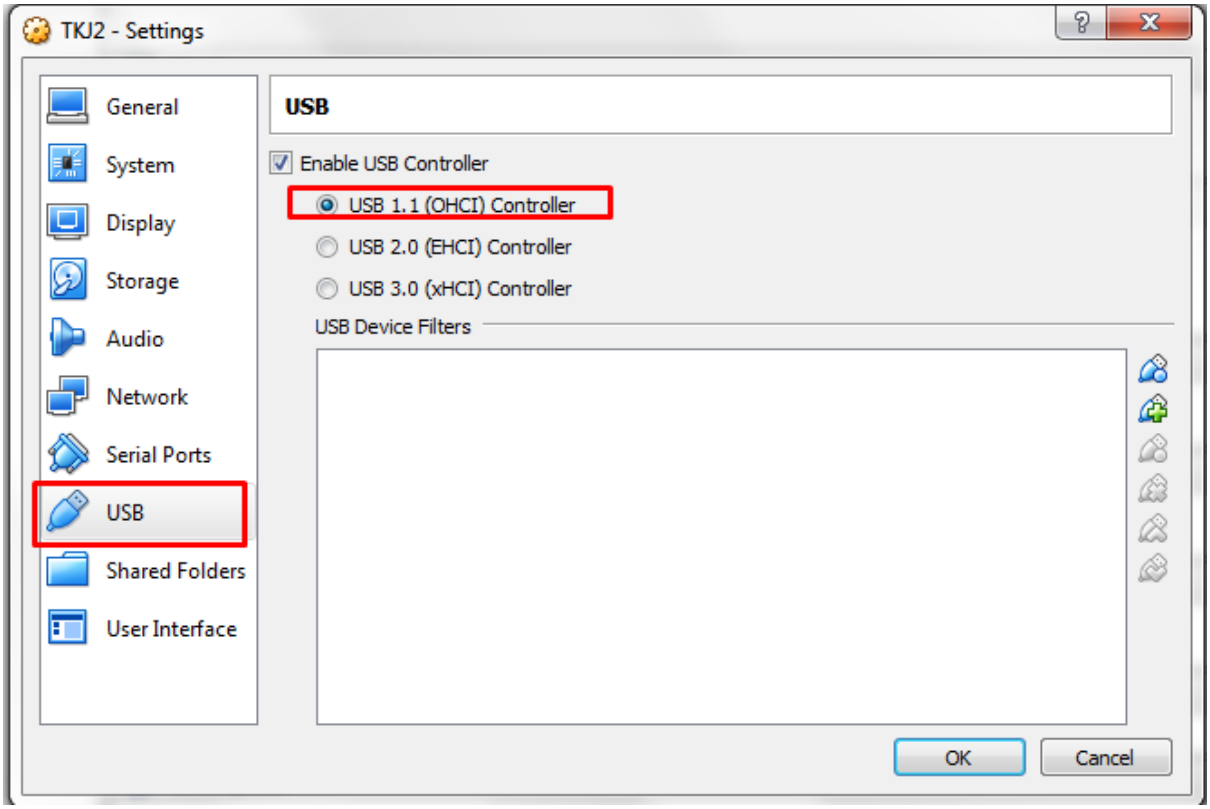


2. Setting adapter networnya seperti berikut:
   a. Adapter 1 = host only
   b. Adapter 2 = NAT

   Set cable connected



Jika masih ada error invalid setting, setting USB menjadi bersi 1.1



3. Login dengan root atau user tkj dan password tekaje
4. Setelah login sebagai root, ketik perintah: `ip link show`
   Perhatikan interface yang muncul
5. Tambahkan DVD 3, kemudian mount
   ```
   #apt-cdrom add
   #apt-cdrom update
   ```
6. Konfigurasi jaringan:
   ```
   #nano /etc/network/interfaces
   ```
7. Konfigurasi IP:
   ```
   auto enp0s3
        iface enp0s3 inet static
        address 192.168.100.1
        netmask 255.255.255.128
   auto enp0s3:2
        iface enp0s3:2 inet static
        address 192.168.(no-absen).1
        netmask 255.255.255.0
   auto enp0s8
        iface enp0s8 inet dhcp
   ```
   simpan ctrl+x lalu y (enter)

```
  GNU nano 3.2                          /etc/network/interfaces

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto enp0s3
        iface enp0s3 inet static
        address 192.168.100.1
        netmask 255.255.255.128

auto enp0s3:2
        iface enp0s3:2 inet static
        address 192.168.40.1
        netmask 255.255.255.0

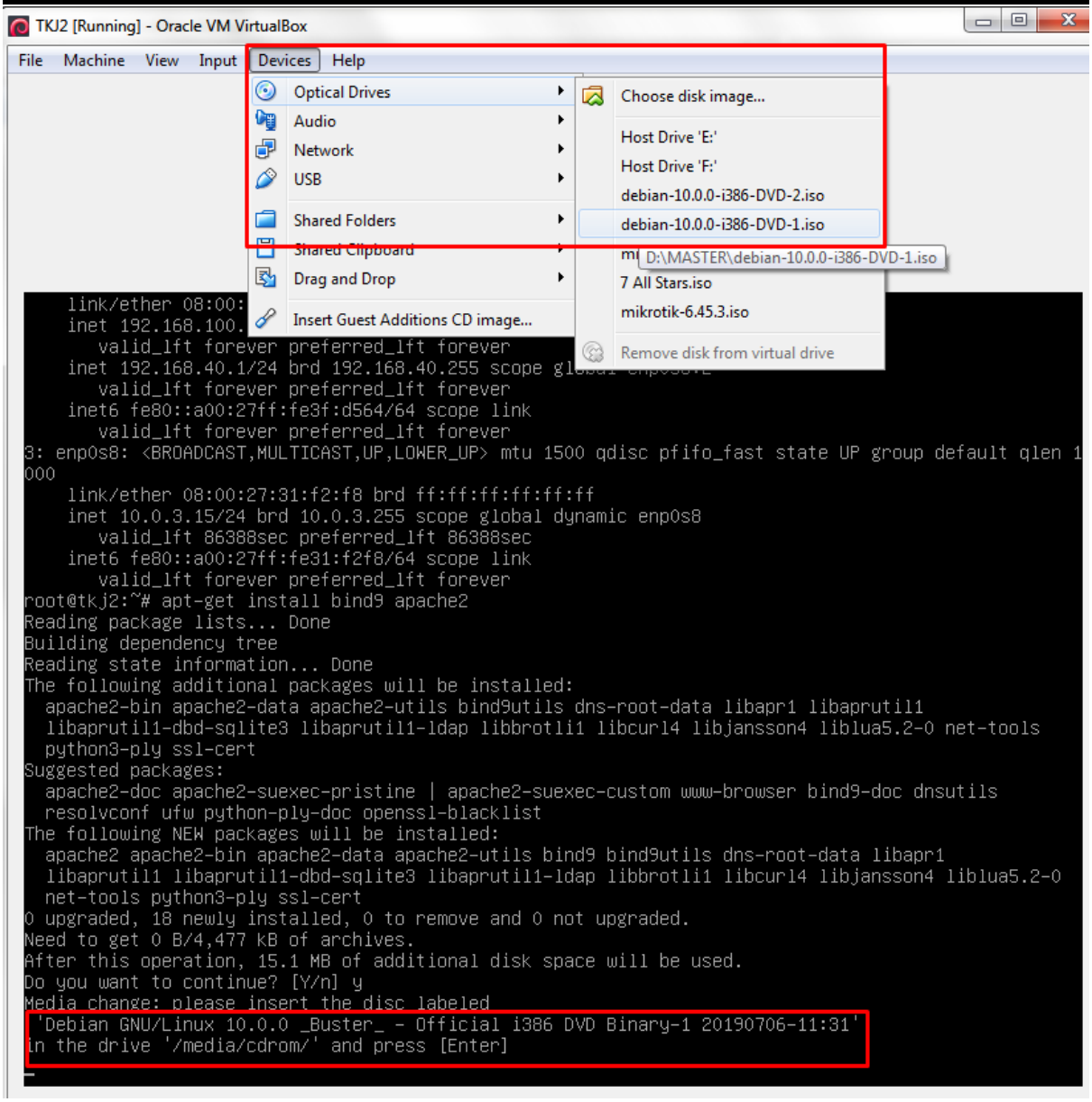
auto enp0s8
        iface enp0s8 inet dhcp
```

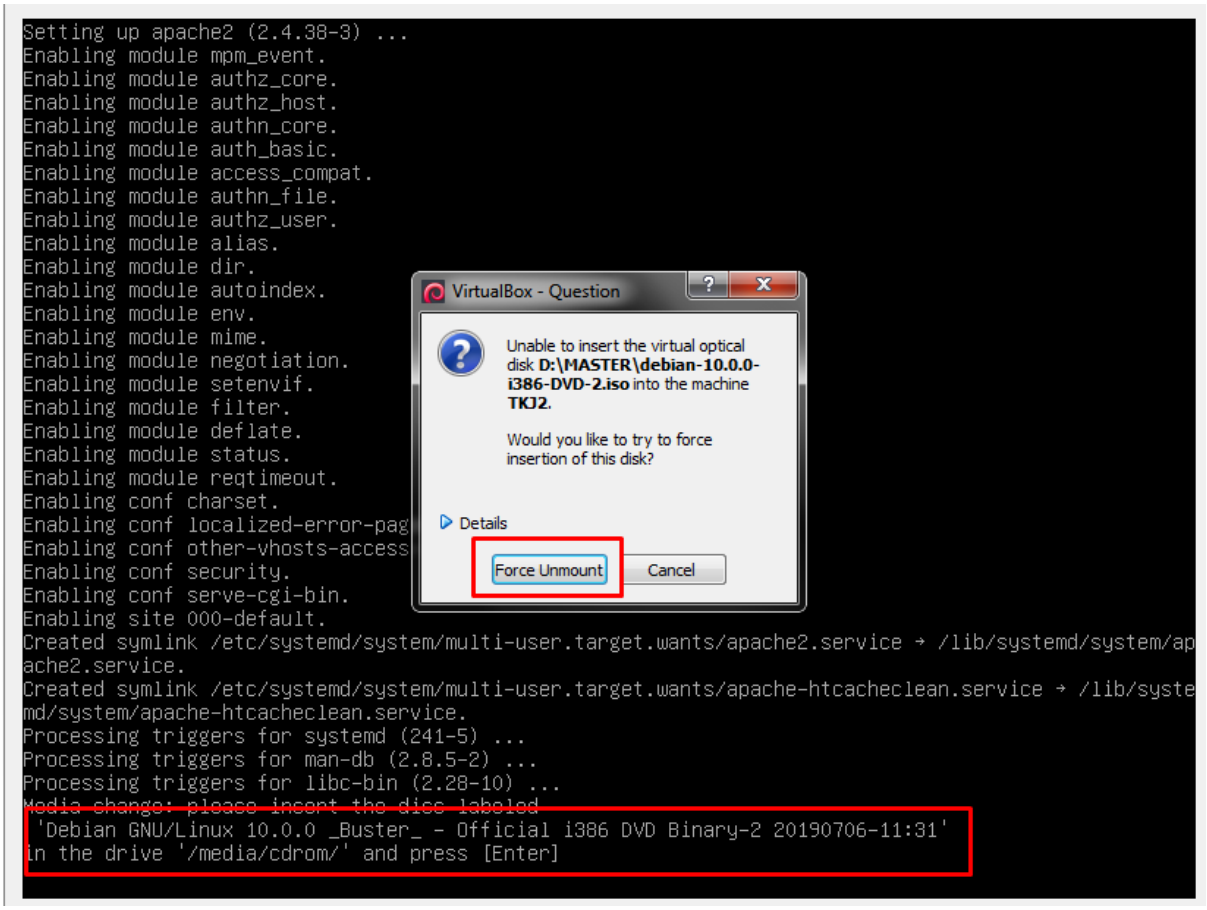
8. Restart
   `#etc/init.d/networking restart (enter)`
   `#ip a atau ip addr show (enter)`

```
root@tkj2:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1
000
    link/ether 08:00:27:3f:d5:64 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.1/25 brd 192.168.100.127 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet 192.168.40.1/24 brd 192.168.40.255 scope global enp0s3:2
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe3f:d564/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1
000
    link/ether 08:00:27:31:f2:f8 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic enp0s8
       valid_lft 86388sec preferred_lft 86388sec
    inet6 fe80::a00:27ff:fe31:f2f8/64 scope link
       valid_lft forever preferred_lft forever
```

9. Install aplikasi DNS server dan Web Server menggunakan paket BIND9 dan apache2
   `#apt-get install bind9 apache2 (enter)` → cara lama atau
   `#apt install bind9 apache2 (enter)` → cara baru
   Jika menggunakan cara baru, terdapat progress ketika installasi berjalan
   TEKAN Y (ENTER)
   INSERT DVD 1 (ENTER)
   INSERT DVD 2 (KLIK FORCE ANMOUNT) (enter)
   bisa gunakan repository ataupun menggunakan DVD.SEBAGAI CONTOH DISINI MENGGUNKAN DVD.

```
root@tkj2:~# apt-get install bind9 apache2
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils bind9utils dns-root-data libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libbrotli1 libcurl4 libjansson4 liblua5.2-0 net-tools
  python3-ply ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser bind9-doc dnsutils
  resolvconf ufw python-ply-doc openssl-blacklist
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils bind9 bind9utils dns-root-data libapr1
  libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap libbrotli1 libcurl4 libjansson4 liblua5.2-0
  net-tools python3-ply ssl-cert
0 upgraded, 18 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/4,477 kB of archives.
After this operation, 15.1 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```





KONFIGURASI DNS SERVER

10. Masuk ke direktori bind
    ```
    #cd /etc/bind
    ```
11. Konfigurasi file default zones
    ```
    #nano named.conf.default-zones (enter)
    Ketik atau tambahkan zones di bagian paling bawah isikan
    nama sesuai nama kalian
    zone "sekolah.sch.id" {
          type master;
          file "/etc/bind/db.sekolah.sch.id";
    };
    zone "(nama).net" {
          type master;
          file "/etc/bind/db.(nama).net";
    };
    (Save)
    ```

```
 GNU nano 3.2                    named.conf.default-zones                    Modified

        file "/etc/bind/db.127";
};

zone "0.in-addr.arpa" {
        type master;
        file "/etc/bind/db.0";
};

zone "255.in-addr.arpa" {
        type master;
        file "/etc/bind/db.255";
};

zone "sekolah.sch.id"{
        type master;
        file "/etc/bind/db.sekolah.sch.id";
};
zone "tkj2.net"{
        type master;
        file "/etc/bind/db.tkj2.net";
};
```

12. Kopi db.local menjadi db.sekolah.sch.id
    `#cp db.local db.sekolah.sch.id`
13. Kopi db.local menjadi db.tkj2.net
    `#cp db.local db.tkj2.net`

```
root@tkj2:/etc/bind# cp db.local db.sekolah.sch.id
root@tkj2:/etc/bind# cp db.local db.tkj2.net
```

Kemudian cek dengan perintah `ls`

```
root@tkj2:/etc/bind# ls
bind.keys  db.255    db.sekolah.sch.id  named.conf.default-zones  rndc.key
db.0       db.empty  db.tkj2.net        named.conf.local          zones.rfc1918
db.127     db.local  named.conf         named.conf.options
```

14. Edit isi dari file db.sekolah.sch.id
    `#nano db.sekolah.sch.id`
    Tekan ctrl+
    Ketik `localhost` untuk mencari kata untuk di replace dan enter

```
Search (to replace): localhost
^G Get Help      M-C Case Sens      M-B Backwards      ^P Older
^C Cancel        M-R Regexp         ^R No Replace      ^N Newer
```

Ketik `sekolah.sch.id` untuk mereplace dari `localhost` ke `sekolah.sch.id`, maka secara otomatis localhost akan terganti dengan `sekolah.sch.id`

```
Replace with: sekolah.sch.id
^G Get Help                ^P Older                ^Y First Line
^C Cancel                  ^N Newer                ^V Last Line
```

Pilih A untuk mereplace secara keseluruhan / ALL
Ganti dibawa NS sekolah.sch.id menjadi

```
@     IN    NS    sekolah.sch.id.
      IN    A     192.168.100.1
ns    IN    A     192.168.100.1
www   IN    A     192.168.100.1
ftp   IN    A     192.168.100.1
```

```
  GNU nano 3.2                          db.sekolah.sch.id

;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     sekolah.sch.id. root.sekolah.sch.id. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      sekolah.sch.id.
        IN      A       192.168.100.1
ns      IN      A       192.168.100.1
www     IN      A       192.168.100.1
ftp     IN      A       192.168.100.1
```

simpan ctrl+x lalu y (enter)

15. Edit file db.tkj2.net

    `#nano db.tkj2.net`

    Lakukan hal yang sama seperti sebelum nya

    Tekan ctrl+

    Ketik `localhost` untuk mencari kata untuk di replace dan enter

```
Search (to replace): localhost
^G Get Help  M-C Case Sens   M-B Backwards  ^P Older   ^Y First Line
^C Cancel    M-R Regexp      ^R No Replace  ^N Newer   ^V Last Line
```

Ketik `tkj.net` untuk mereplace dari `localhost` ke `tkj.net`, maka secara otomatis localhost akan terganti dengan `tkj.net`

```
Replace with: tkj2.net
^G Get Help                 ^P Older   ^Y First Line
^C Cancel                   ^N Newer   ^V Last Line
```

Pilih A untuk mereplace secara keseluruhan / ALL

Ganti di bawah NS tkj2.net menjadi

```
@       IN      NS      tkj2.net.
        IN      A       192.168.40.1
ns      IN      A       192.168.40.1
www     IN      A       192.168.40.1
```

```
  GNU nano 3.2                          db.tkj2.net                          Modified

;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     tkj2.net. root.tkj2.net. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      tkj2.net.
        IN      A       192.168.40.1
ns      IN      A       192.168.40.1
www     IN      A       192.168.40.1
```

simpan ctrl+x lalu y (enter)

16. Restart jaringan
    `#/etc/init.d/networking restart`
17. Restart bind9
    `#/etc/init.d/bind9 restart`

```
root@tkj2:/etc/bind# /etc/init.d/networking restart
[ ok ] Restarting networking (via systemctl): networking.service.
root@tkj2:/etc/bind# /etc/init.d/bind9 restart
[ ok ] Restarting bind9 (via systemctl): bind9.service.
```

    Jika failed cek kembali konfigurasi default-zones, db.sekolah.sch.id, db.tkj2.net. perhatikan penulisan titik koma, tanda petik, titik, kurung buka dan tutup dls.
18. Edit file resolv.conf, tambahkan nameserver 192.168.100.1 di bagian paling atas
    `#nano /etc/resolv.conf`
19. Cek apakah domain sudah berhasil ditambahkan
    `#host -l sekolah.sch.id`
    `#host -l tkj2.net`
    `#nslookup sekolah.sch.id`

```
root@tkj2:/etc/bind# host -l sekolah.sch.id
sekolah.sch.id name server sekolah.sch.id.
sekolah.sch.id has address 192.168.100.1
ftp.sekolah.sch.id has address 192.168.100.1
ns.sekolah.sch.id has address 192.168.100.1
www.sekolah.sch.id has address 192.168.100.1
root@tkj2:/etc/bind# host -l tkj2.net
tkj2.net name server tkj2.net.
tkj2.net has address 192.168.40.1
ns.tkj2.net has address 192.168.40.1
www.tkj2.net has address 192.168.40.1
root@tkj2:/etc/bind# nslookup sekolah.sch.id
Server:         192.168.100.1
Address:        192.168.100.1#53

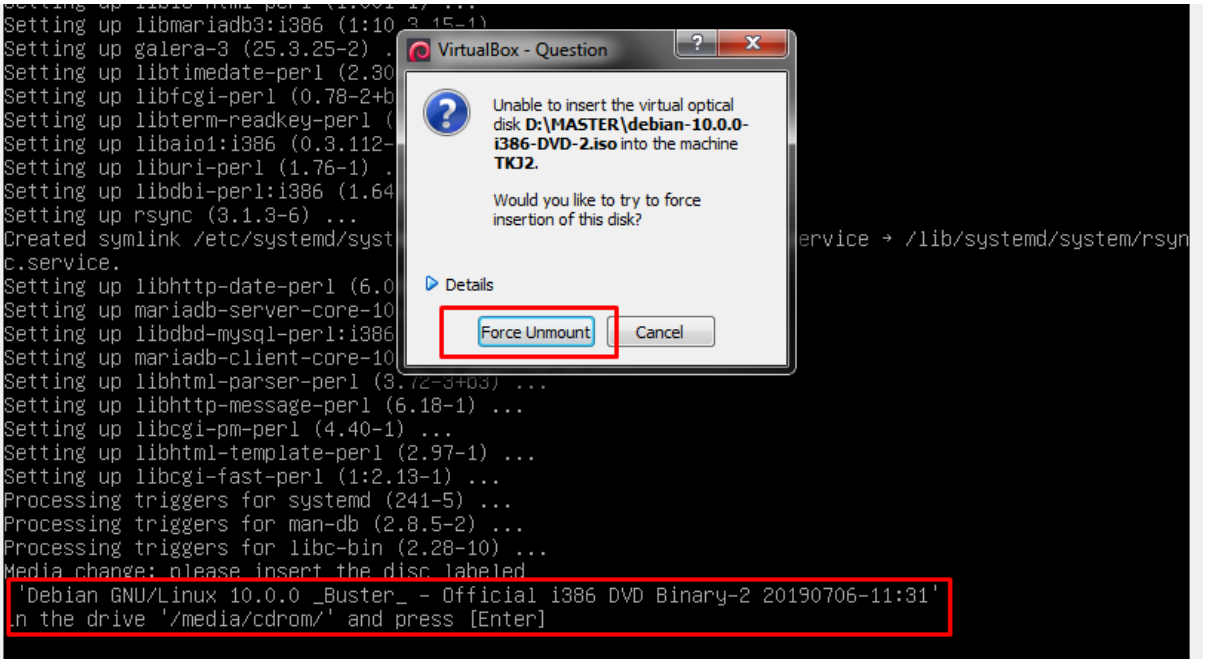
Name:   sekolah.sch.id
Address: 192.168.100.1

root@tkj2:/etc/bind# ns
nsec3hash  nsenter    nslookup   nstat      nsupdate
root@tkj2:/etc/bind# nslookup tkj2.net
Server:         192.168.100.1
Address:        192.168.100.1#53

Name:   tkj2.net
Address: 192.168.40.1
```

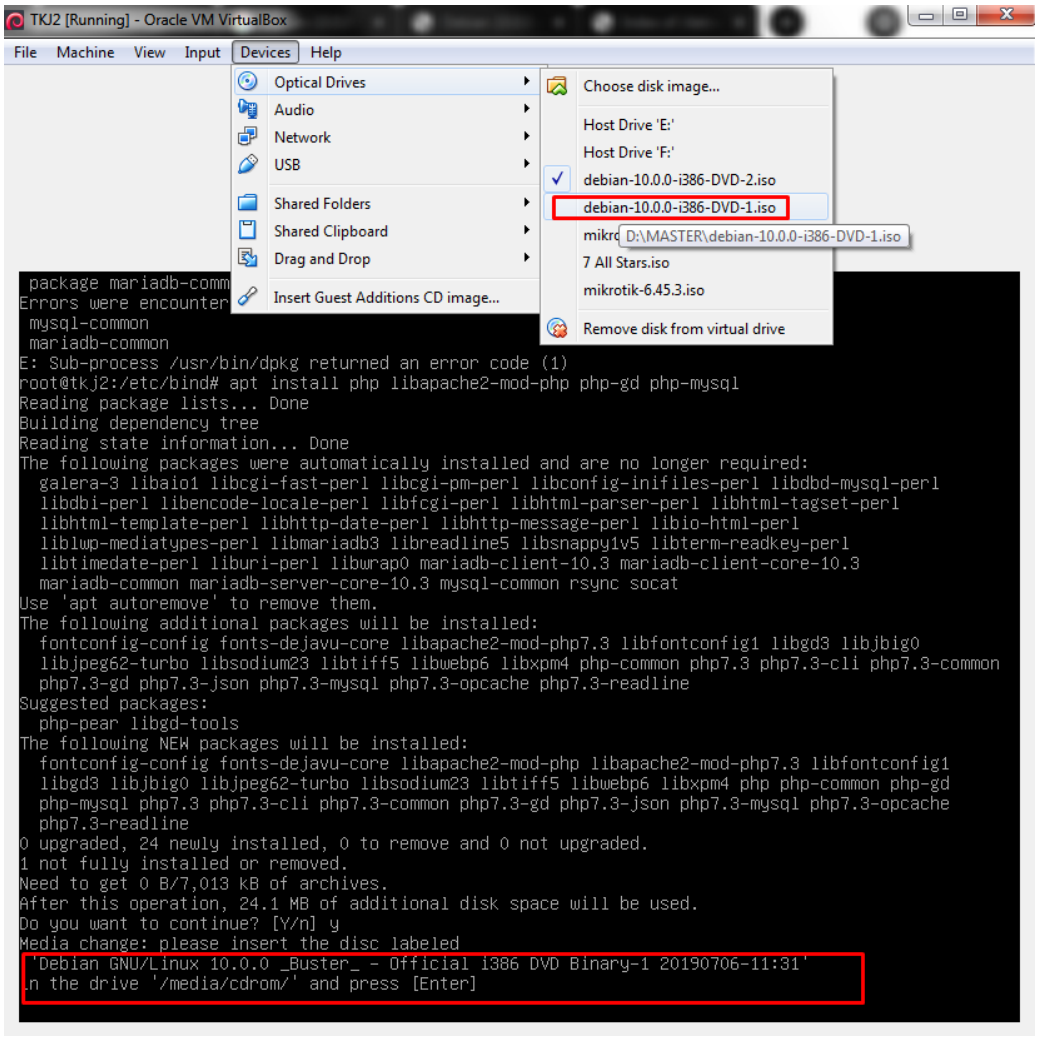
KONFIGURASI WEB SERVER

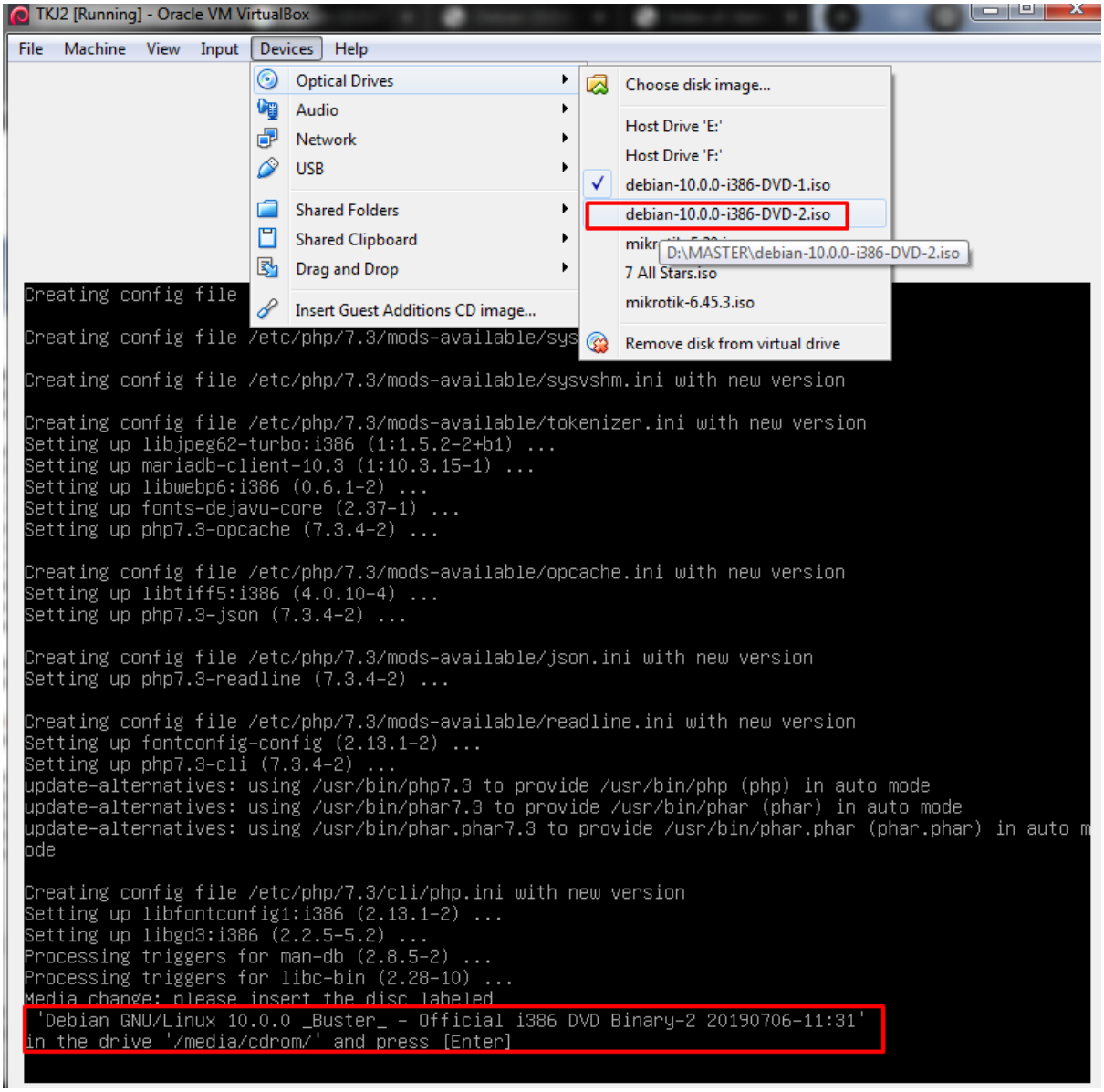
20. Install DBMS Mysql (mariaDB) versi terbaru
    `#apt install mariadb-server`
    Masukkan DVD sesuai perintah



21. Install PHP

```
# apt install php libapache2-mod-php php-gd php-mysql
```





22. Konfigurasi VirtualHost pada webserver

```
#mkdir /var/www/html/sekolah.sch.id
# mkdir /var/www/html/tkj2.net
# cd /etc/apache2/sites-available/
# cp 000-default.conf sekolah.sch.id.conf
# cp 000-default.conf tkj2.net.conf
```

```
root@tkj2:/etc/bind# mkdir /var/www/html/sekolah.sch.id
root@tkj2:/etc/bind# mkdir /var/www/html/tkj2.net
root@tkj2:/etc/bind# cd /etc/apache2/sites-available/
root@tkj2:/etc/apache2/sites-available# cp 000-default.conf sekolah.sch.id.conf
root@tkj2:/etc/apache2/sites-available# cp 000-default.conf tkj2.net.conf
```

23. Edit file sekolah.sch.id.conf

```
# nano sekolah.sch.id.conf
(hapus tanda # di depan ServerName)
ServerName www.sekolah.sch.id

ServerAdmin admin@sekolah.sch.id
DocumentRoot /var/www/html/sekolah.sch.id
```

(save)

```
 GNU nano 3.2                          sekolah.sch.id.conf

<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        ServerName sekolah.sch.id

        ServerAdmin admin@sekolah.sch.id
        DocumentRoot /var/www/html/sekolah.sch.id

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf
</VirtualHost>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet
```

24. edit file tkj2.net.conf

```
# nano tkj2.net.conf (enter)
ServerName www.tkj2.net

ServerAdmin webmaster@tkj2.net
DocumentRoot /var/www/html/tkj2.net
```

Save

```
 GNU nano 3.2                            tkj2.net.conf

<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        ServerName www.tkj2.net

        ServerAdmin webmaster@tkj2.net
        DocumentRoot /var/www/html/tkj2.net

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf
</VirtualHost>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet
```

25. restart servis webserver

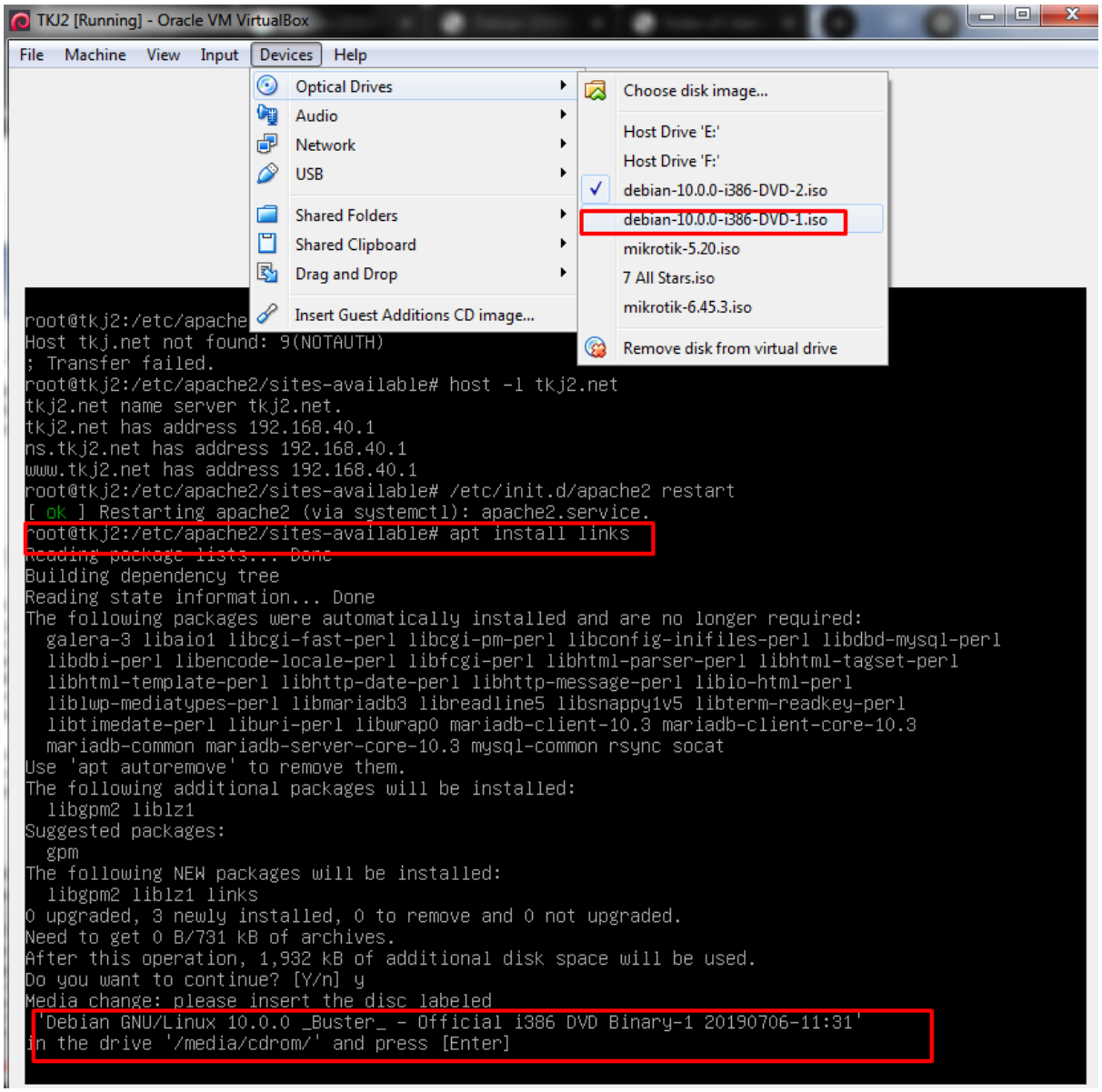
```
# /ect/init.d/apache2 restart
```

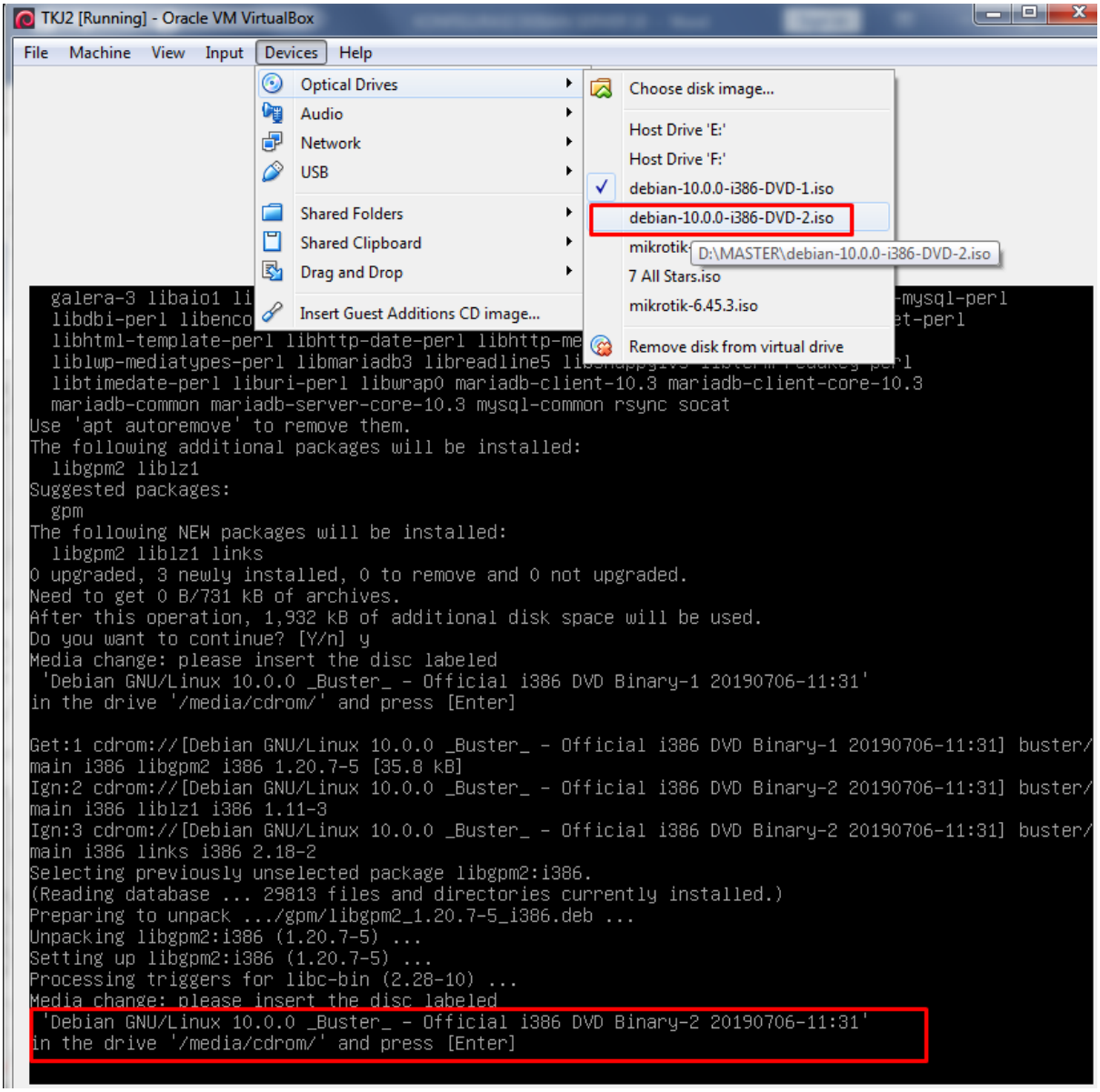
```
root@tkj2:/etc/apache2/sites-available# /etc/init.d/apache2 restart
[ ok ] Restarting apache2 (via systemctl): apache2.service.
```

UJI COBA DNS DAN WEBSERVER

26. install Links (CLI browser) dengan perintah : fungsinya untuk melihat atau menampilkan browser dalam mode CLI

```
# apt install links
```

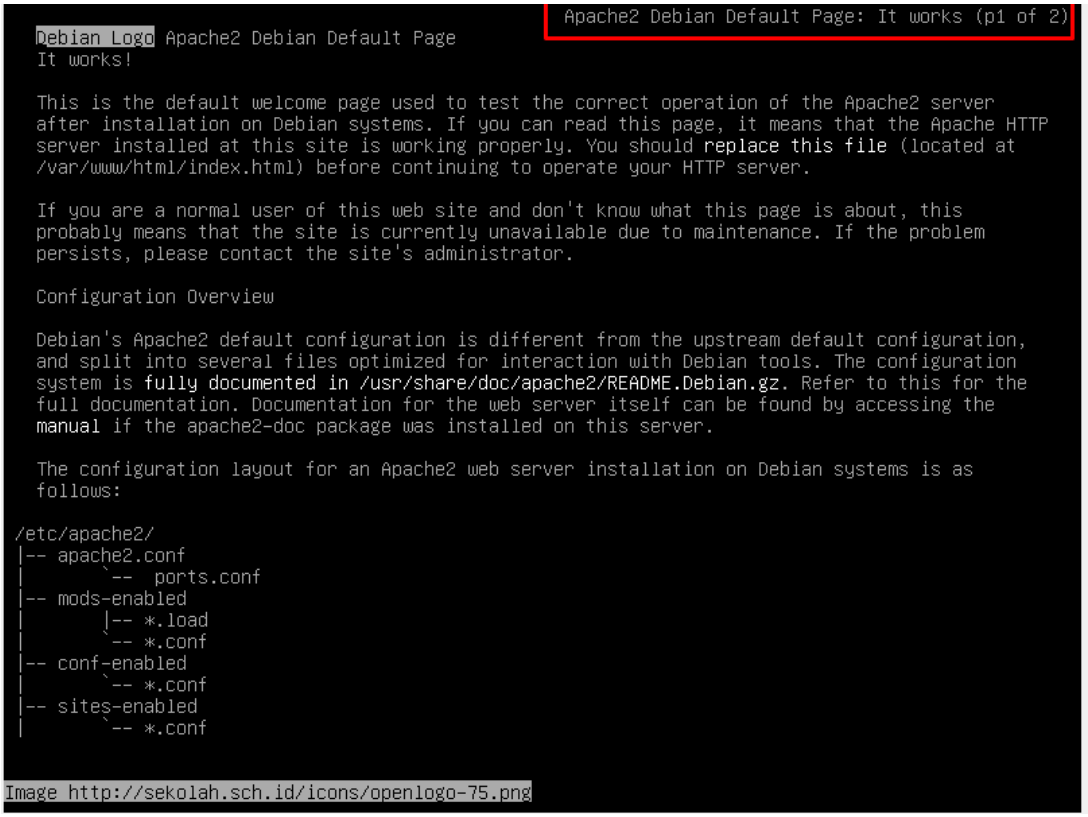




27. akses ke sekolah.sch.id menggunakan links dengan perintah:

    ```
    # links sekolah.sch.id
    ```

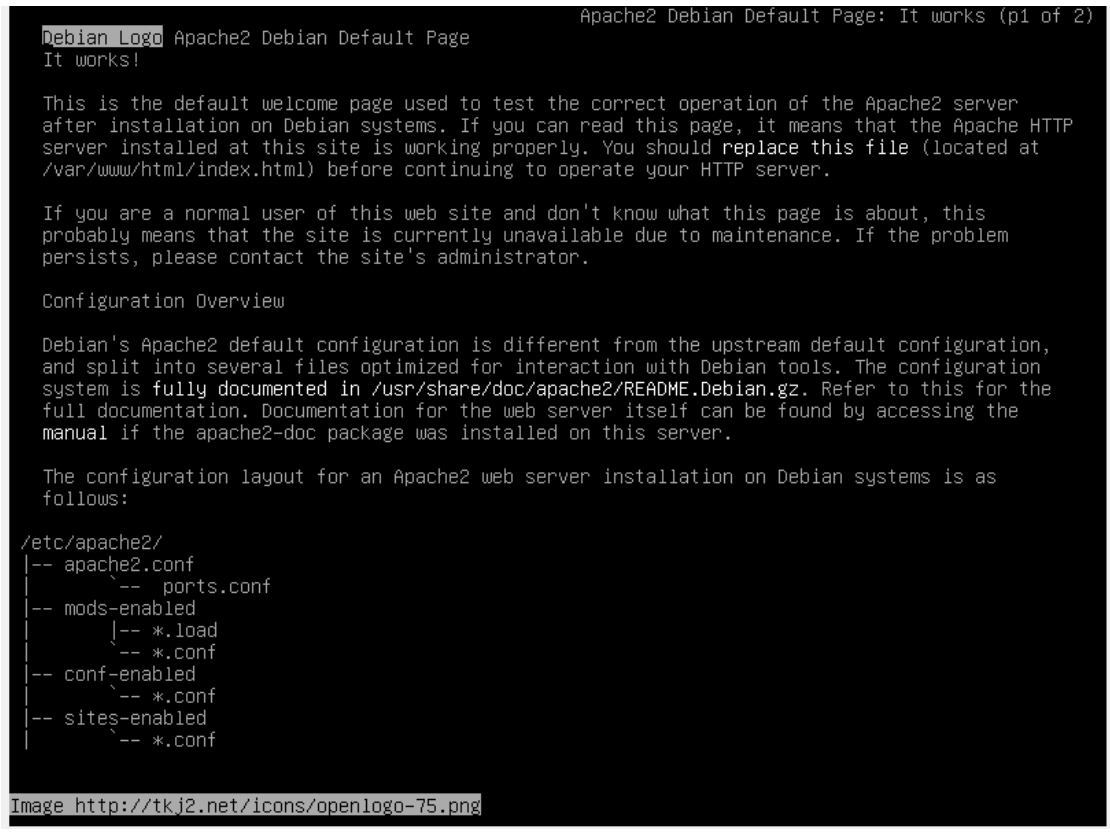
    (Jika keluar tampilan default page Apache maka konfigurasi Apache Web server telah berhasil)



Untuk keluat tekan ctrl+c

28. Keluar links dengan menekan ctrl+c lalu akses tkj2.net

```
# links tkj2.net
```



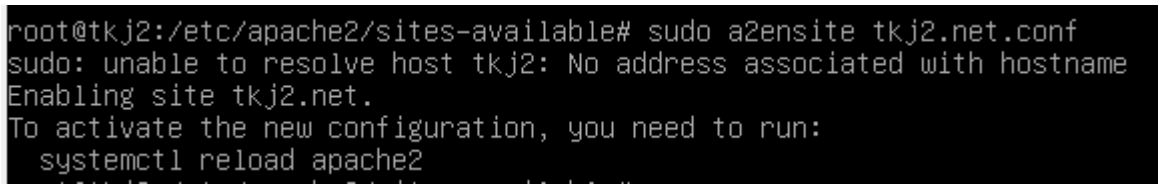
INSTALLASI DAN KONFIGURASI WEB CMS WODRPRESS

29. Install program unzip dan php extension untuk Web CMS
    ```
    # apt install unzip php-curl
    # apt install sudo
    ```
30. Masuk ke direktori sites-available apache2
    ```
    # cd /etc/apache2/sites-available/
    ```
31. Edit file virtualhost tkj2.net
    ```
    # nano tkj2.net.conf
    ```
    Hapus www pada bagian ServerName menjadi tkj2.net

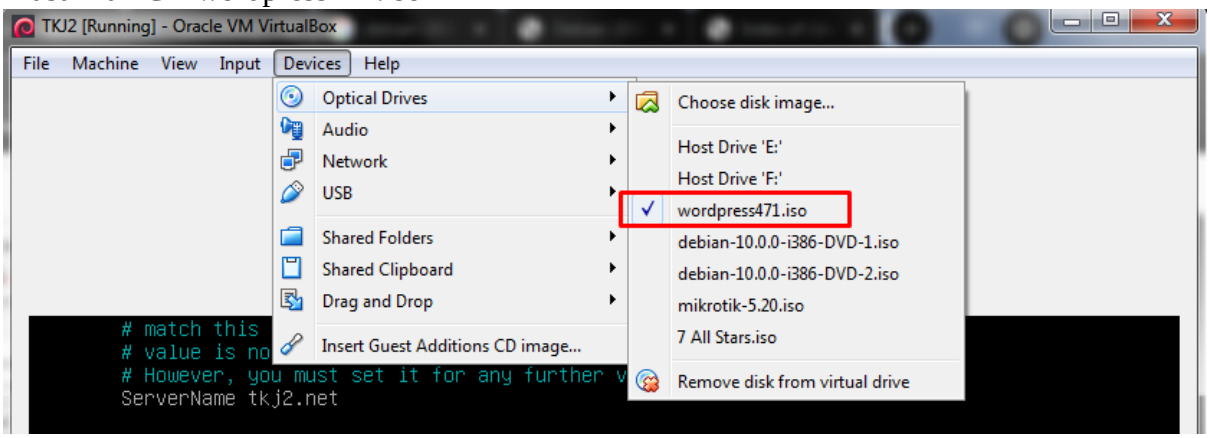


    Save

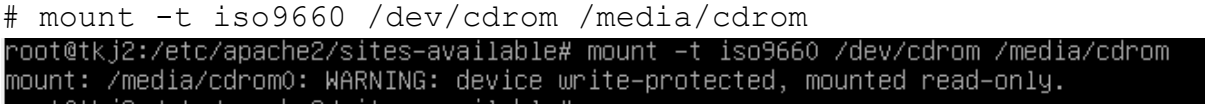
32. Masukkan perintah
    ```
    # sudo a2ensite tkj2.net.conf
    ```



    Kemudian restart apache2
    ```
    # /etc/init.d/apache2 restart
    ```
33. Masukkan CD wordpress471.iso



    Mount ke /media/cdrom
    ```
    # mount -t iso9660 /dev/cdrom /media/cdrom
    ```



    Kemudian ls

```
root@tkj2:/etc/apache2/sites-available# cd /media/cdrom
root@tkj2:/media/cdrom# ls
index.html  wordpress-4.7.1.zip
```

34. Copy file wordpress-4.7.1.zip dan index.html ke folder /tmp

```
# cd /media/cdrom
# cp wordpress-4.7.1.zip /var/www/html/tkj2.net/
# cp index.html /var/www/html/tkj2.net/
# cd /var/www/html/tkj2.net/
# unzip wordpress-4.7.1.zip
# chmod -R 755 /var/www/html/tkj2.net/wordpress/
# chown -R www-data.www-data
/var/www/html/tkj2.net/wordpress
```

```
root@tkj2:/media/cdrom# cp wordpress-4.7.1.zip /var/www/html/tkj2.net/
root@tkj2:/media/cdrom# cp index.html /var/www/html/tkj2.net/
root@tkj2:/media/cdrom# cd /var/www/html/tkj2.net/
root@tkj2:/var/www/html/tkj2.net# unzip wordpress-4.7.1.zip
```

```
root@tkj2:/var/www/html/tkj2.net# chmod -R 755 /var/www/html/tkj2.net/wordpress
root@tkj2:/var/www/html/tkj2.net# chown -R www-data.www-data /var/www/html/tkj2.net/wordpress
```

35. Install database server

```
#apt install mariadb-server
```

36. Buat database untuk wordpress

```
# mysql -u root -p
(ketikkan password tekaje)
> CREATE USER 'tekaje'@'localhost' IDENTIFIED BY
'tekaje';
> CREATE DATABASE wpdb; (enter)
> GRANT ALL ON `wpdb`.* TO `tekaje`@`localhost`;
> FLUSH PRIVILEGES;
> exit (enter)
# /etc/init.d/mysql restart (enter)
```

```
MariaDB [(none)]> CREATE USER 'tekaje'@'localhost' IDENTIFIED BY 'tekaje';
Query OK, 0 rows affected (0.002 sec)

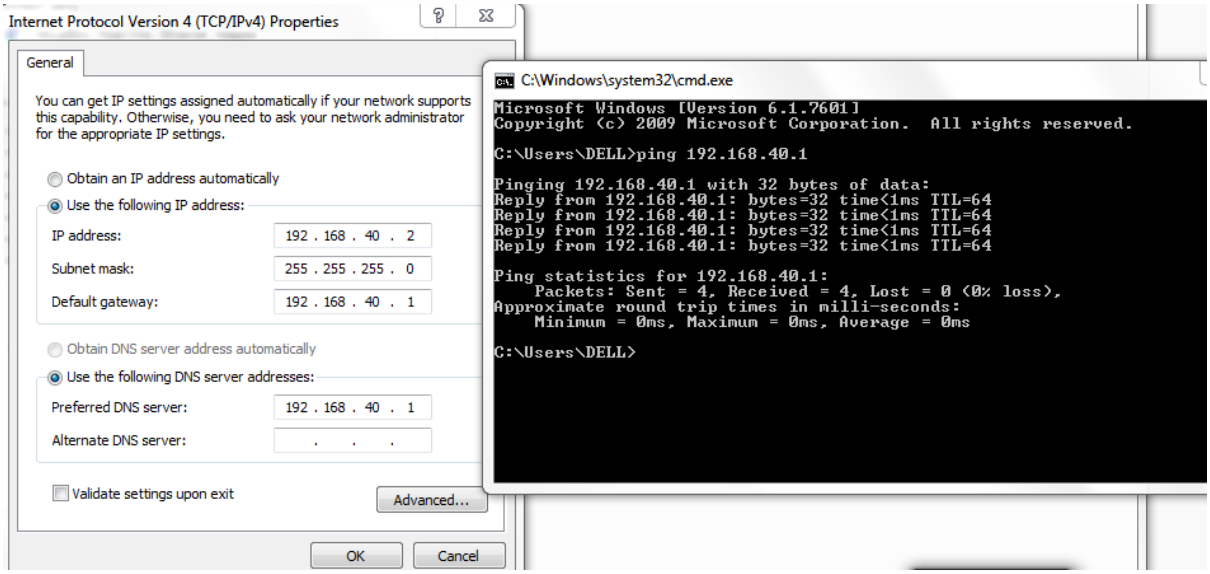
MariaDB [(none)]> CREATE DATABASE wpdb;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> GRANT ALL ON `wpdb`.* TO `tekaje`@`localhost`;
Query OK, 0 rows affected (0.001 sec)

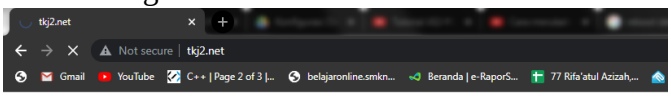
MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> exit_
```

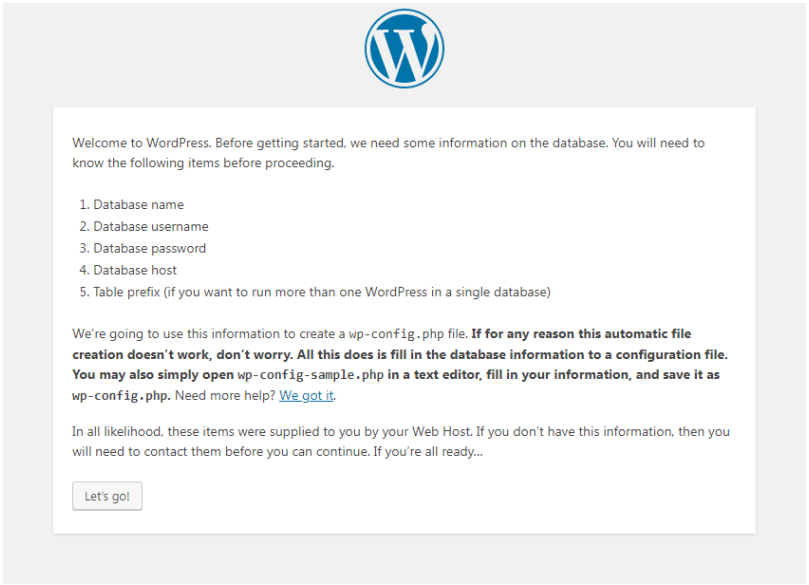
37. Ganti IP address pada windows menjadi
    Address : 192.168.(no-absen).2
    Subnet : 255.255.255.0
    Gateway : 192.168.(no-absen).1
    DNS : 192.168.(no-absen).1



38. Buka windows explorer masuk ke direktori
    `C:\Windows\System32\drivers\etc`
    Ubah status Read Only file hosts
    Buka file `hosts` dengan program notepad++
    Tambahkan
    `192.168.40.1 tkj2.net`
    Save as administrator
39. Buka Browser ketik alamat 192.168. 40.1
    lalu akses ke alamat www.tkj2.net
    pilih bahasa lalu klik “Continue”
    klik "Let's go"

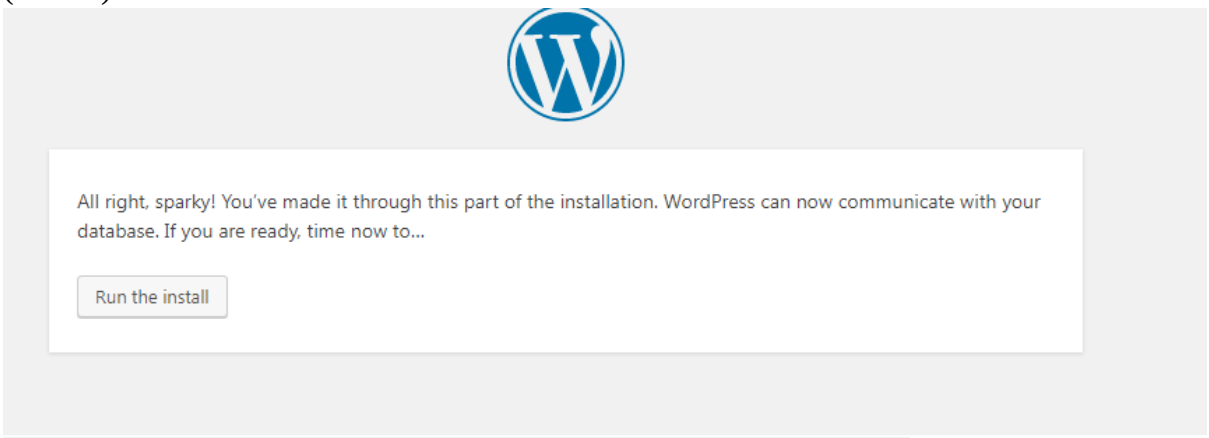


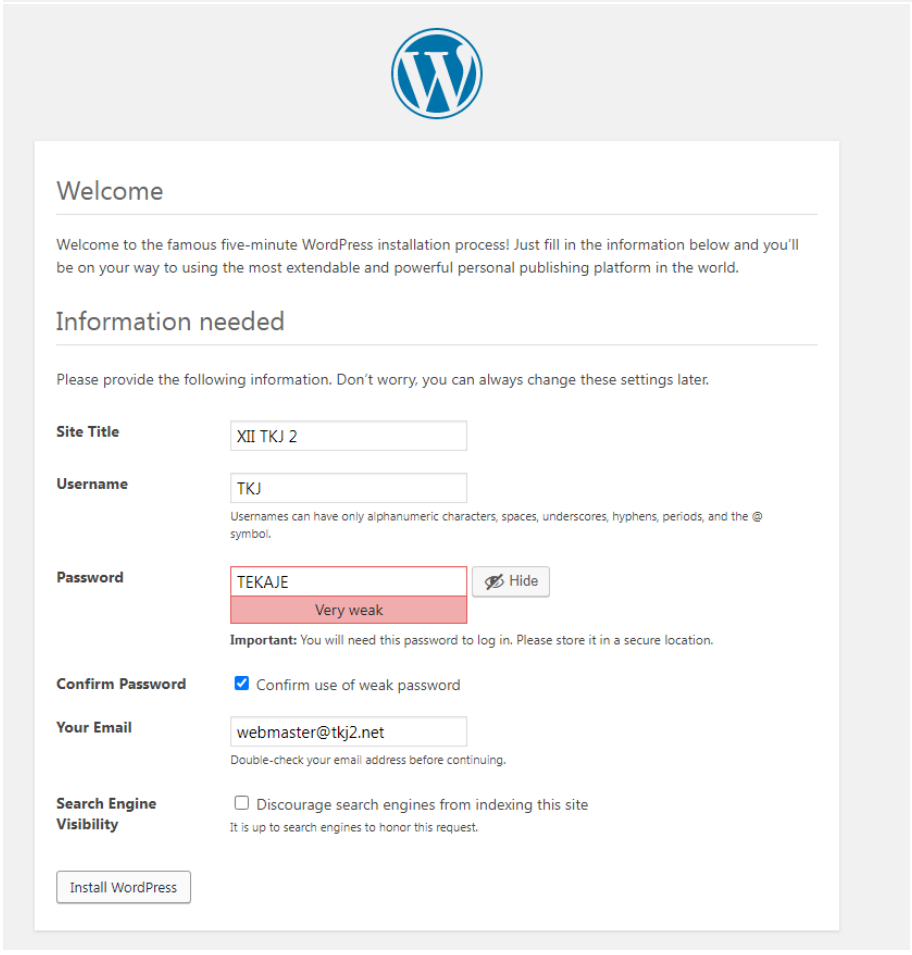
Loading wordpress...

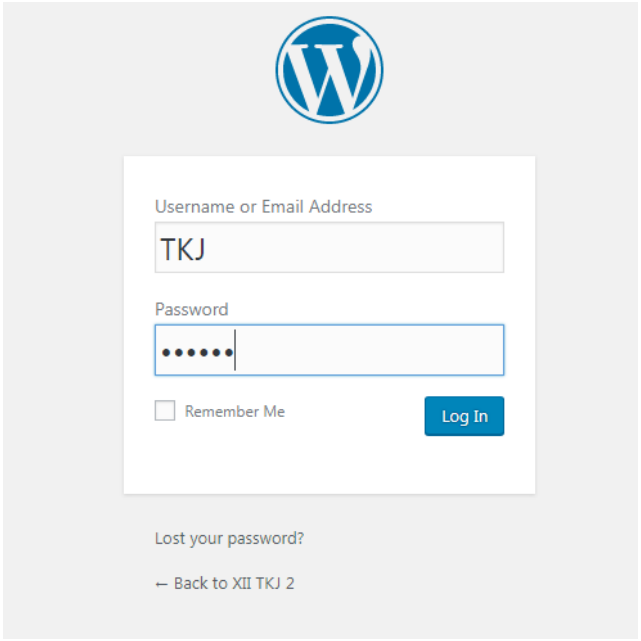


40. Masukkan data sesuai dengan yang sudah dikonfigurasi
    `Database name : wpdb`

Username : tekaje
Password : tekaje
Database Host : localhost
Table Prefix : wp_
(submit)







KLIK PADA VIEW SITE

